Advanced Mathematics Applications

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Contents

Preface

These notes follow the course *Advanced Mathematics Applications*, taught by Lyle Noakes at the University of Western Australia in Semester 2 of 2026. The unit teaches advanced techniques of calculus, including multivariable calculus, Green's, Stoke's and divergence theorem, complex analysis, ordinary and partial differential equations, and Sturm-Liouville theory.

0.1 More detailed overview of the subject.

These notes are about something but with more detail.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every *or* for all
$\exists$	There exists
$\implies$	Implies
$\in$	In *or* is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not *or* without
$\varnothing$	Empty set
$\sim$	Relates to
$\mapsto$	Maps to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$\lvert n \rvert$	The absolute value of n
$\square$	It has been proven

Part I

Essentials of Multivariable Calculus

Chapter 1

Vectors, Vector Valued Functions, and Functions of Several Variables

1.1 The Vector Space $\mathbb{R}^n$

A VECTOR SPACE V on $\mathbb{F}$ is the set V of objects named *vectors* and a field[1] $\mathbb{F}$ of objects name *scalars*, with two operations *vector addition* and *scalar multiplication* defined:

1. Vector Addition: $+ \colon V \times V \to V$
2. Scalar Multiplication: $\cdot \colon \mathbb{F} \times V \to V$

That satisfies the axioms of a vector space. Namely that vector addition and scalar multiplication are both commutative, associative, and distributive, and that there exists additive and multiplicative identities and inverses.

THE VECTOR SPACE $\mathbb{R}^n$ is the set $V = \{(x_0, x_1, \dots, x_n) : x_i \in \mathbb{R}\}$ over the real numbers. Elements of V are said to be n-tuples of real numbers. Vector addition is defined to be the coordinate-wise addition of elements of V:

$$(x_0, x_1, \dots, x_n) + (x'_0, x'_1, \dots, x'_n) = (x_0 + x'_0, x_1 + x'_1, \dots, x_n + x'_n)$$

with the usual arithmetic inherited from the real numbers. Scalar multiplication is defined as the distributive multiplication of each coordinate by some $\alpha \in \mathbb{R}$:

$$\alpha \cdot (x_0, x_1, \dots, x_n) = (\alpha \cdot x_0, \alpha \cdot x_1, \dots, \alpha \cdot x_n)$$

In two and three dimensions, n-tuples of numbers can be thought of as a quantity with *magnitude and direction*, and arrow from the origin to (x, y, z), however to not lose generality in higher dimensions it is best just to think of them as ordered lists of real numbers, with arithmetic defined above.

[1] refer: Algebra

THE DOT PRODUCT of two vectors is defined:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{n=1}^{n} a_i b_i$$

for two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. It is also known as the scalar or inner product. An alternative formulation of the dot product is:

$$\mathbf{a} \cdot \mathbf{b} = ||\mathbf{a}||\ ||\mathbf{b}|| \cos\theta$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, and $||\mathbf{a}||$ is the magnitude, or length, of $\mathbf{a}$. Thus, two orthogonal vectors will have a dot product of 0 (since $\cos(\pi/2) = 0$) and two parallel vectors will have a dot product equal to the product of their magnitudes. Notice

$$\mathbf{a} \cdot \mathbf{a} = ||\mathbf{a}||^2 \implies ||\mathbf{a}|| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$$

The dot product can be thought of geometrically as a measure of how much of $\mathbf{a}$ *goes in* the direction of $\mathbf{b}$. A handy application of this fact is determining the x, y, and z components of a vector $\mathbf{a}$, by taking its dot product with the standard basis vectors[2] $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$, respectively.

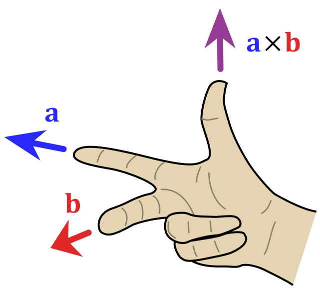

Figure 1.1: The right hand rule is the convention used for the positive direction $\mathbf{n}$ in $\mathbf{a} \times \mathbf{b}$.

THE CROSS PRODUCT of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ is defined:

$$\mathbf{a} \times \mathbf{b} = (||\mathbf{a}||\ ||\mathbf{b}|| \sin\theta) \cdot \mathbf{n}$$

where θ is the angle between them and $\mathbf{n}$ is a unit vector perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$, whose direction is given by the right hand rule (fig. 1.1). A vector perpendicular to a plane is said to be *normal* to that plane.

An easy way to calculate the cross product is to build the 3×3 matrix with first row $[\mathbf{i}\ \mathbf{j}\ \mathbf{k}]$, second row $[\mathbf{a}]$, and third row $[\mathbf{b}]$, and take its determinant.

[2] $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2)\mathbf{i} + (a_3 b_1 - a_1 b_3)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}$$

Once you know a normal vector to a plane (usually by taking the cross product of two vectors in that plane) and one point in the plane, it is easy to determine an equation for that plane, since the dot product of *any* vector in the plane with the normal vector will be 0.

$$\mathbf{n} \cdot (x, y, z) = n_x x + n_y y + n_z z + d = 0$$

Simply plug in the (x, y, z) that you know to determine the correct value of d.

1.2 Vector Valued Functions

A FUNCTION f is said to *vector valued* if it is a mapping $\mathbb{R} \to \mathbb{R}^n$. In other words, vector valued function take a real number and assign a vector.

$$\mathbf{r}(t) = (x_1(t), x_2(t), \ldots, x_n(t))$$

The functions $x_1(t), \ldots, x_n(t)$ are called the *coordinate functions* of $\mathbf{r}$.

A common example of a vector valued function is $\mathbf{r}(t) = (t, f(t))$ where $f : \mathbb{R} \to \mathbb{R}$ and $t \in [a, b] : a, b \in \mathbb{R}$. This function describes a *parametrisation* of f and points from the origin to every point of f as t varies in its domain, as if the tip were moving along f. It is helpful to imagine t as representing time. This will be covered in more detail in Chapter 7.

1.3 Functions of Several Variables

A FUNCTION f is a *multivariable* function if it is a mapping from $\mathbb{R}^n \to \mathbb{R}$. These functions take a vector and assign to it a real number. The coordinates of the input vector can be thought of as several independent variables.

$$f(x_1, x_2, \ldots, x_n) = n$$

where n is some combination of the variables $x_1, \ldots, x_n$. Just like we can plot a function of one variable $f : D \to R$ by defining the line $l = \{(x, f(x)) : x \in D\}$ as the set of all points that satisfy $(x, f(x))$, we can define an $n+1$ dimensional surface of a function $f : D \subseteq \mathbb{R}^n \to R \subseteq \mathbb{R}$ of n variables by

$$S = \{(x_1, x_2, \ldots, x_n, f(x_1, x_2, \ldots, x_n)) : x_1, \ldots, x_n \in D\}$$

Allowing an $n+1$ dimensional alien to understand its plot. The only applications where this is useful to humans is to plot $z = f(x, y)$ or every point (x, y, z) that satisfies $f(x, y, z) = k$ for some $k \in \mathbb{R}$ in three dimensions.

Fields are mappings $\mathbb{R}^n \to \mathbb{R}^m$, usually $\mathbb{R}^3 \to \mathbb{R}^m$. They can be thought of as assigning to every point in space a vector in $\mathbb{R}^m$. A common example is a force or electromagnetic field. A scalar field is a special multivariable function of three variables that assigns to every point in space a number, like temperature or pressure. These are covered in Chapter 8.

Chapter 2

Limits and Continuity

2.1 Functions of One Variable

A function f approaches a limit l near a if for every $\epsilon > 0$, there exists some $\delta > 0$, such that if $0 < |x-a| < \delta$, then $|f(x) - l| < \epsilon$.

This is a precise way of writing that if a function approaches a value l near a, then no matter how small of an interval we draw around l, we will always be able to draw an interval around a, where $f(x)$ is within that interval of l. It will become clear in the examples why this definition matches the intuitive understanding of what a limit should be.

Consider $f(x) = x^2$ (fig. 2.1). Obviously it approaches 0 near 0 (and a^2 near a), but how would we show that? We have to decide how to ensure that $|x^2 - a^2| < \epsilon$ whenever $|x-a| < \delta$. We want

$$|x^2 - a^2| = |x-a| \cdot |x+a| < \epsilon$$

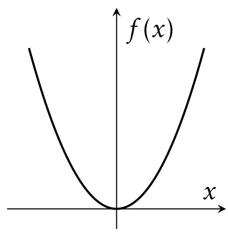

Figure 2.1: $f(x) = x^2$.

The factor $|x+a|$ is obviously the problem, but as long as we know *an* upper bound M for it, we can just ensure that $|x-a| < 1/M$. If we take $|x+a| < 1$, then

$$|x+a| = |(x-a) + 2a| \leq |x-a| + |2a| < 1 + 2|a|$$

Now, we just need

$$|x-a| < \frac{\epsilon}{1+2|a|}$$

for $|x^2 - a^2| < \epsilon$. So, let $\epsilon > 0$, choose $\delta = \min\left(1, \frac{\epsilon}{1+2|a|}\right)$. Then $\delta \leq 1$, so $|x-a| < 1$ and $|x+a| < 1 + 2|a|$. Consider $|x^2 - a^2| = |x-a| \cdot |x+a|$. Substituting our bounds, $|x^2 - a^2| = \delta \cdot (1 + 2|a|) \leq \frac{\epsilon}{1+2|a|} \cdot (1 + 2|a|) = \epsilon$.

So for every ϵ, there is an interval around a, equal to $a \pm \min\left(1, \frac{\epsilon}{1+2|a|}\right)$ that will ensure $|f(x) - l| < \epsilon$.

Consider $f(x) = \sin(1/x)$. (fig. 2.2)

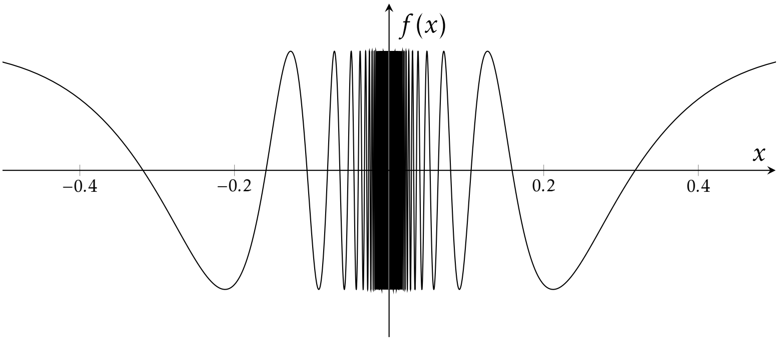

Figure 2.2: $f(x) = \sin(1/x)$.

This function oscillates wildly as $x \to 0$ and so does not approach any value at $x = 0$. The intuition for this fact is that no matter how small we make an interval containing 0, we will always be able to choose a large enough natural number n, such that there is a point $a = \frac{2}{\pi(1+4n)}$ where $\sin a = 1$, and a point $b = \frac{2}{\pi(3+4n)}$ where $\sin b = -1$.

Assume for contradiction that $\lim_{x\to 0} \sin 1/x = l$. Let $\epsilon = 1/2$ and $\delta > 0$. Now choose a sufficiently large natural number n, such that $a = \frac{2}{\pi(1+4n)} < \delta$. Then, there will also be the point $b = \frac{2}{\pi(3+4n)} < a < \delta$. Then,

$$
\begin{aligned}
|f(a) - l| = |1 - l| &< \epsilon = \frac{1}{2} \\
|f(b) - l| = |-1 - l| &< \epsilon = \frac{1}{2}
\end{aligned}
$$

By the triangle inequality

$$|1+1| = |(1-l)+(1+l)| \leq |1-l|+|1+l| < \frac{1}{2}+\frac{1}{2} = 1 \implies 2 < 1$$

Which is a contradiction. So our original assumption that the limit exists and equals l must be wrong. $\square$

A SLIGHTLY MORE COMPLICATED EXAMPLE. fig. 2.3

$$f(x) = \begin{cases} 0 & x \text{ irrational, } 0 < x < 1 \\ \frac{1}{q} & x = \frac{p}{q} \text{ in lowest terms, } 0 < x < 1 \end{cases}$$

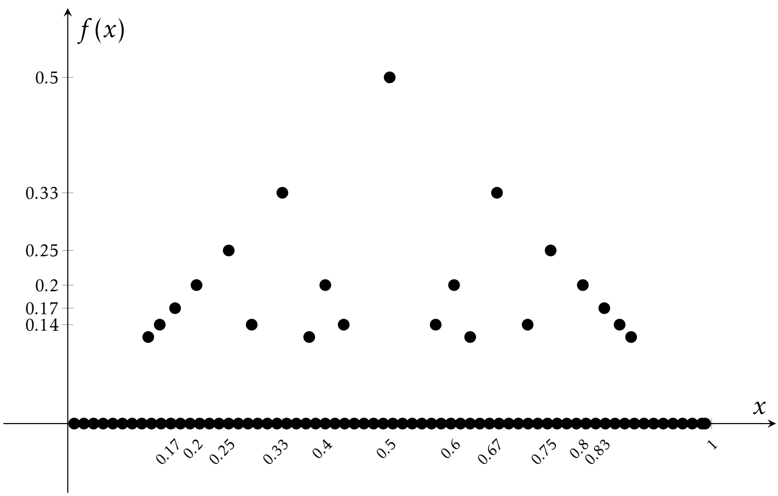

Figure 2.3: $f(x)$.

Perhaps counterintuitively at the moment, this function approaches 0 at a. Recall the definition of a limit, the function f *approaches* l at a if for every $\epsilon > 0$, $\exists \delta > 0$, with $0 < |x-a| < \delta$, and $|f(x)-l| < \epsilon$. We don't actually care what $f(a)$ is, since the difference between x and a will always be greater than 0, the value at $f(a)$ is irrelevant.

Let $\epsilon > 0$ be given. Choose a sufficiently large natural number n such that

$1/n < \epsilon$. The only numbers x where $|f(x) - 0| < \epsilon$ could be false are

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \ldots, \frac{1}{n}, \ldots, \frac{n-1}{n}$$

the set of which we can call S.

However many of these numbers there are, there are only finitely many, and so one of them is closest to a. That is, $|p/q - a|$ is smallest for one specific $p/q \in S$. If a is in S, then just consider all the elements of $S \neq a$. If we chose δ to be this smallest distance then for all x satisfying $0 < |x - a| < \delta$, x is guaranteed not to be in S, and therefore $|f(x) - 0| < \epsilon$, which completes our proof. □

There is no need to give δ in terms of ϵ, since there is nowhere in the definition $\ldots\forall\epsilon, \exists\delta\ldots$ that says each ϵ needs a *unique* δ.

2.1.1 Divergence to, and Limits at, Infinity

The definition of a limit can be modified to handle divergence to infinity. A function l approaches ∞ near a if for every $M > 0$, there exists a δ such that, whenever $|x - a| < \delta$, $f(x) > M$. No matter how large M is, there will always be an interval we can draw around a, such that $f(x) > M$.

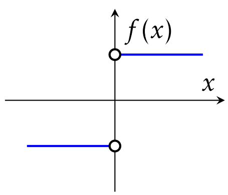

Figure 2.4: The derivative of the absolute value function.

Limits at infinity have a similar modified definition. A function f approaches a limit l near ∞ if for every $\epsilon > 0$, there exists an $M > 0$ such that if $x > M$, $|f(x) - l| < \epsilon$. For every interval around the limit we draw, there is always a point on the number line that ensures we are within that interval.

2.1.2 One Sided Limits

One-sided limits are exactly the same as the full limit, except that x is only considered to be approaching a from the left or the right. The full limit only exists if both one-sided limits equal, but it is possible for either one-sided limit to exist independently.

For example, define (fig. 2.4)

$$f(x) := \frac{d|x|}{dx} = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases}$$

If we examine what f does as $x \to 0$ from the left, its quite obvious that it approaches -1, and 1 from the right. We say that

$$\lim_{x\to 0^-} f(x) = -1$$
$$\lim_{x\to 0^+} f(x) = 1$$

but the full limit:

$$\lim_{x\to 0} f(x)$$

is not defined.

2.2 Vector Valued Functions

We get these for free. Just define

$$\lim_{t\to a} \mathbf{r}(t) := \left(\lim_{t\to a} x_1(t), \lim_{t\to a} x_2(t), \ldots, \lim_{t\to a} x_n(t)\right)$$

If the limits of the coordinate functions exist, then they can be evaluated using the same techniques as single variable functions

2.3 Techniques for Evaluating Limits

The limit laws allow for breaking up complicated limits in several trivial to solve limits.

1. $\lim_{x\to a}(f+g)(x) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$
2. $\lim_{x\to a} f(x) \cdot g(x) = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$
3. $\lim_{x\to a} c \cdot f(x) = c \cdot \lim_{x\to a} f(x)$
4. $\lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$

5. If $\lim_{x \to a} g(x) = l$ and f is continuous at l, then $\lim_{x \to a} f(g(x)) = f(l)$

THE SQUEEZE THEOREM states that if $g(x) \leq f(x) \leq h(x)$ and $\lim_{x \to a} g(x) = l = \lim_{x \to a} h(x)$, then $\lim_{x \to a} f(x) = l$.

L'HÔPITAL'S RULE states that if f and g are differentiable, and $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0$ or $\pm\infty$, then $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \lim_{x \to a} \dfrac{f'(x)}{g'(x)}$.

Proofs of these assertions are a bit beyond the scope of these notes.

2.4 Limits of Multivariable Functions

FOR LIMITS of multivariable functions, one must consider every possible path to a. Just like with functions of one variable, where we require the left and right sided limits to equal, for multivariable functions, we require all paths to be equal. For example:

$$\lim_{(x,y) \to (0,0)} f(x,y) = \frac{xy}{x^2 + y^2}$$

Consider the path $x = y$:

$$\lim_{(x,y) \to (x,x)} f(x,y) = \lim_{(x,y) \to (x,x)} \frac{x^2}{x^2 + x^2} = \frac{1}{2}$$

Now, consider the path $x = 0$:

$$\lim_{(x,y) \to (0,y)} f(x,y) = \lim_{(x,y) \to (0,y)} \frac{0}{y^2} = 0 \neq \frac{1}{2}$$

Thus, the limit does not exist.

The technique for solving limits of multivariable functions is to investigate a few paths to observe their behaviour. If the limit exists then a symbolic proof must be given to cover all possible paths. Otherwise, one example will suffice. Evaluating the limit uses similar techniques to single variable limits.

Consider

$$\lim_{(x,y) \to (0,0)} \frac{x^2 + x + xy + y}{x + y}$$

Naively evaluating $(0,0)$ produces an indeterminate $\dfrac{0}{0}$, but L'Hôpital's rule is not defined for multivariable functions. However if we factorise the numerator:

$$\lim_{(x,y)\to(0,0)} \frac{x^2+x+xy+y}{x+y} = \lim_{(x,y)\to(0,0)} \frac{(x+y)(x+1)}{x+y} = \lim_{(x,y)\to(0,0)} x+1 = 1$$

We did not make any assumptions about the path taken by x and y to $(0,0)$, and so this symbolic proof remains true for every possible path.

2.5 Continuity

A function is continuous if $\lim_{x\to a} f(x) = f(a)$. Common continuous functions include

1. All polynomials $p_o + p_1 x + p_2 x^2 + \cdots + p_n x^n$.
2. All rational functions $R(x) = \dfrac{P(x)}{Q(x)}$ except where $Q(x) = 0$.
3. $\sin x$ and $\cos x$.
4. e^x and $\ln x$ (on its domain $x \in (0,\infty)$).
5. x^α for $\alpha \in \mathbb{R}$ and $x > 0$.
6. Linear combinations and products of continuous functions are continuous.
7. Ratios of continuous functions are continuous, provided the denominator is not 0.
8. $f(g(x))$ is continuous when $g(x)$ is continuous, provided the codomain of g is the domain of f.

Chapter 3

Derivatives

3.1 Single Variable Functions

For a function f, define its *derivative* at a:

$$f'(a) := \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

If this limit exists, the function is called *differentiable*. Define

$$f'(x) := \{(a, f'(a)) : f'(a) \text{ exists.}\}$$

to be all the points $(a, f'(a))$ where $f'(a)$ exists, called the derivative function.

This number, $f'(a)$ measures the slope of a tangent line to f at a, sometimes called the instantaneous rate of change or, when $f(x)$ represents motion, *velocity*. The equation for a tangent line T to f at a (fig. 3.1):

$$T(x) = f'(a)(x-a) + f(a)$$

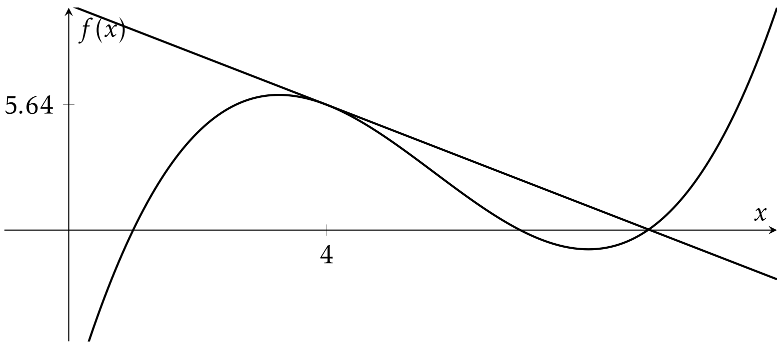

Figure 3.1: The derivative measures the slope of a tangent line.

If a function is differentiable at a, then it is continuous at a. *Proof:*

$$\begin{aligned}
\lim_{h\to 0} f(a+h)-f(a) &= \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}\cdot h \\
&= \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}\cdot \lim_{h\to 0} h \\
&= f'(a)\cdot 0 \\
&= 0 \\
\implies \lim_{h\to 0} f(a+h)-f(a) &= 0 \\
\implies \lim_{x\to a} f(x) &= f(a)
\end{aligned}$$

□

Note the inverse is certainly not true. The absolute value function is continuous everywhere, but not differentiable at 0 (fig. 2.4). In fact, we can conceive of a function that is continuous everywhere but differentiable nowhere. Consider a function like fig. 3.2

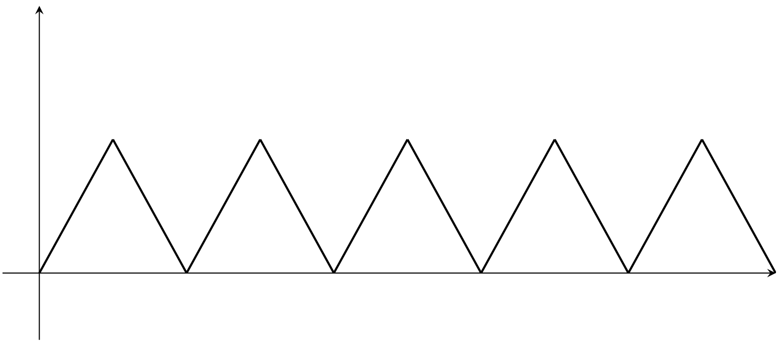

Figure 3.2: A 'sawtooth' like function is not differentiable at its points. In the limit as the number of points goes to ∞, differentiability goes to 0.

One may wish to know more about a function and its derivatives. For example, if $d(t)$ represents distance from an origin w.r.t time, then $d'(t)$ represents the velocity of the object w.r.t time. If $v(t) = d'(t)$, then $v'(t)$ represents the acceleration w.r.t time. We say that acceleration is the *second derivative* of $d(t)$, written $d''(t)$. Define

$$f^{(n)} := (((f')')')^{\cdots}$$

To be the n^{th} derivative of f. This collection of derivatives are known as higher-order derivatives.

3.1.1 Notation

For single variable functions Newton's f', f'', f''', etc. is unambiguous. However, when partial derivatives of functions of several variables are introduced, it will become necessary to specify which variable the function is being differentiated with respect to. For this reason, Leibniz's operator notation is commonly used.

$$\frac{df}{dx} = f'(x)$$
$$\frac{d^2 f}{dx^2} = f''(x)$$
$$\frac{d^n f}{dx^n} = f^{(n)}(x)$$

3.2 Differentiating

Once the derivatives of the elementary functions have been obtained, there are some crucial rules for finding the derivatives of their products, sums, quotients and compositions. See the table for derivatives of elementary functions.

Assuming f and g are differentiable, and defined on the prerequisite domains:

1. $[f \pm g]' = f' \pm g'$
2. $\forall c \in \mathbb{R}, [c \cdot f]' = c \cdot f'$
3. $[fg]' = fg' + gf'$
4. $\left[\dfrac{f}{g}\right]' = \dfrac{1}{g^2}(gf' - fg')$
5. $[f \circ g]' = (f' \circ g) \cdot g'$

3.2.1 Inverse Functions

Let f be a bijective function on its domain. Then there exists a function g called the *inverse* of f, such that $(g \circ f)(x) = x$. The inverse function of f is commonly notated f^{-1}. This is not to be confused with f's multiplicative inverse $1/f(x)$, which will be denoted like that.

Assuming f is differentiable and invertible,

$$(f^{-1})' = \frac{1}{f' \circ f^{-1}}$$

For example:

$$(\ln x)' = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

3.2.2 Table of Derivatives

Function	Derivative
c	0
x^n	nx^{n-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\csc x$	$-\csc x \cot x$
$\sec x$	$\sec x \tan x$
$\cot x$	$-\csc^2 x$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\csc^{-1} x$	$-\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$

Function	Derivative
$\sec^{-1} x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$
$\cot^{-1} x$	$-\frac{1}{1+x^2}$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$
$\operatorname{csch} x$	$-\operatorname{csch} x \coth x$
$\operatorname{sech} x$	$-\operatorname{sech} x \tanh x$
$\coth x$	$-\operatorname{csch}^2 x$
e^x	e^x
$a^x \quad (a > 0)$	$a^x \ln a$
x^x	$x^x(\ln x + 1)$
$\ln x$	$1/x$
$\log_a x \ (a > 0,\ a \neq 1)$	$1/(x \ln a)$
$\frac{1}{x^n + a}$	$-\frac{nx^{n-1}}{(x^n+a)^2}$

3.3 Vector Valued Functions

THE DERIVATIVE of a vector valued function $\mathbf{r}(t)$ is

$$\mathbf{r}'(t) = (x_1'(t), x_2'(t), \dots, x_n'(t))$$

where $x_1(t), \dots, x_n(t)$ are the coordinate functions of $\mathbf{r}(t)$.

The same rules apply to differentiating vector valued functions as above, except

1. $[\mathbf{u} \cdot \mathbf{v}]' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$

2. $[\mathbf{u} \times \mathbf{v}]' = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$

RECALL that if $\mathbf{r}(t)$ traces a curve C in space, then $\mathbf{r}'(t)$ will be tangent to C at t. Define

$$\mathbf{T}(t) := \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

to be the unit tangent vector to C at t. Since $\mathbf{r}(t)$ encodes position along a curve C, $\mathbf{r}'(t)$ encodes the velocity at t (including direction). Acceleration is given by $\mathbf{r}''(t)$, with unit acceleration given by $\mathbf{T}'(t)$.

3.4 Multivariable Functions

FOR A FUNCTION $f : \mathbb{R}^n \to \mathbb{R}$, we cannot meaningfully speak about the derivative of f before first deciding which variable we are differentiating with respect to. Define the *partial derivative* of f w.r.t x_n at $\mathbf{c}$:

$$f_{xi}'(\mathbf{c}) = \frac{\partial f(\mathbf{c})}{\partial x_i} = \lim_{h \to 0} \frac{f(c_{x1}, \dots, c_{xi} + h, c_{xn}) - f(\mathbf{c})}{h}$$

This is calculated by treating all other $x_j \in \mathbf{c}$ as constants, and differentiating only w.r.t x_i. They can then be read from derivative tables or calculated using the methods of single variable functions.

If we imagine a path along some surface in $\mathbb{R}^3$ defined by $S = \{(x, y, f(x, y))\}$, fixing $y = b$ and $x = a$:

$$\mathbf{r}_x(x) = (x, b, f(x, b))$$
$$\mathbf{r}_y(y) = (a, y, f(a, y))$$

describe curves of constant y and x respecively along S. Taking their first partial derivatives w.r.t x and y,

$$\mathbf{r}'_x(x) = (1, 0, f'_x(a, b))$$
$$\mathbf{r}'_y(y) = (0, 1, f'_y(a, b))$$

we arrive at two vectors perpendicular to S at $(a, b, f(a, b))$, and thus they form a *tangent plane* P to S. We can now obtain a normal vector to that plane:

$$\mathbf{n} = \mathbf{r}'_x(x) \times \mathbf{r}'_y(y) = (-f'_x(a, b), -f'_y(a, b), 1)$$

Since $\mathbf{n}$ is necessarily perpendicular to P, its dot product with any vector in P is 0, and so it takes the form:

$$-f'_x(a, b)x - f'_y(a, b)y + z = d$$

For some $d \in \mathbb{R}$, which can be determined by using the fact that $(a, b, f(a, b)) \in P$.

3.4.1 Gradient Vector and Direction Derivatives

Let $f : \mathbb{R}^n \to \mathbb{R}$. Define the gradient operator:

$$\nabla = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \ldots, \frac{\partial}{\partial x_n} \right)$$

to be the vector operator in $\mathbb{R}^n$ of each partial derivative w.r.t x_i. Then $\nabla f(\mathbf{c})$, called the gradient vector of f at $\mathbf{c}$ is the n-dimensional vector, normal to a level curve of f at $\mathbf{c}$ that points in the direction of maximum rate of change of f. This provides techniques for determining two properties of curves (if we're lucky!).

The directional derivative of $f(\mathbf{c})$ towards $\mathbf{u}$ is the rate of change of $f(\mathbf{c})$ in the direction of $\mathbf{u}$. It can be calculated by taking the dot product of a unit vector in the direction of $\mathbf{u}$ with $\nabla f(\mathbf{c})$:

$$\text{Directional Derivative} = \nabla f(\mathbf{c}) \cdot \frac{\mathbf{u}}{||\mathbf{u}||}$$

If S is a level surface, i.e. it is of the form $S = \{f(\mathbf{c}) = k\}$, then $\nabla f(\mathbf{c})$ has the same dimension as $\mathbf{c}$, and thus is normal to S. A tangent plane can then be described via the same technique above.

3.4.2 Chain Rule for Multivariable Functions

If f is a function of $x_1, x_2, \ldots, x_n$, and x_i is a function of t, then:

$$\frac{df}{dt} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{dx_i}{dt}$$

Alternatively, if x_i are functions of s and t, then:

$$\frac{\partial f}{\partial s} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial s}$$
$$\frac{\partial f}{\partial t} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial t}$$

3.4.3 The Jacobian Matrix

If $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$, then:

$$\mathbf{F}(x_1, x_2, \ldots, x_n) = \begin{bmatrix} f_1(x_1, x_2, \ldots, x_n) \\ f_2(x_1, x_2, \ldots, x_n) \\ \vdots \\ f_m(x_1, x_2, \ldots, x_n) \end{bmatrix}$$

The derivative of $\mathbf{F}$ is then an $n \times m$ matrix, called the Jacobian matrix made up of all partial derivatives of $\mathbf{F}$:

$$\mathbf{F}' = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

If $(x, y, z) = \mathbf{F}(u, v, w)$ describes a coordinate transformation from (x, y, z) to (u, v, w), then the determinant of the Jacobian matrix, called the Jacobian, is a scaling factor needed when changing coordinate systems in integration, described further in Chapter 7. For example:

$$\begin{aligned} (x, y) &= \mathbf{F}(r, \theta) = (r\cos\theta, r\sin\theta) \\ \implies \mathbf{F}'(r, \theta) &= \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} \\ |\mathbf{F}'| &= r \end{aligned}$$

So,

$$\int_{-r}^{r} \int_{-\sqrt{r-x^2}}^{\sqrt{r-x^2}} 1 dy dx = \int_{0}^{r} \int_{0}^{2\pi} r d\theta dr = \pi r^2$$

3.5 Maxima and Minima

Maxima and minima are, as the name suggests, points of maximum and minimum. A point a is a maximum of f if for every c, $f(c) \leq f(a)$, and vice versa for minima.

A point c is a *critical point* of f if $f'(c) = 0$. A critical point is a maximum if $f''(c) < 0$ and a minimum if $f''(c) > 0$.

For multivariable functions we must instead look at the gradient vector. A point $\mathbf{c}$ is a critical point of a function $f : \mathbb{R}^n \to \mathbb{R}$ if $\nabla f(\mathbf{c})$ doesn't exist, or

$\nabla f(\mathbf{c}) = 0$. Define the Hessian matrix to be the square matrix of n $2nd$ order partial derivatives of f:

$$D^2 f(\mathbf{c}) = \begin{bmatrix} f_{xx} & f_{xy} & \cdots & f_{xn} \\ f_{yx} & f_{yy} & \cdots & f_{yn} \\ \vdots & \vdots & \ddots & \vdots \\ f_{nx} & f_{ny} & \cdots & f_{nn} \end{bmatrix}$$

Then if this matrix exists and $\nabla f(\mathbf{c}) = 0$ (or dne), the determinant $|D^2 f(\mathbf{c})|$, denoted $D_f(\mathbf{c})$ provides the information for the multivariable equivalent of the second derivative test.

1. If $D_f(\mathbf{c}) > 0$ and $f_{xx} > 0$ then f has a local minimum at $\mathbf{c}$
2. If $D_f(\mathbf{c}) > 0$ and $f_{xx} < 0$ then f has a local maximum at $\mathbf{c}$
3. If $D_f(\mathbf{c}) < 0$ then f has a saddle point at $\mathbf{c}$

3.5.1 Lagrange Multipliers

Let $f(\mathbf{x})$ and $g(\mathbf{x})$ be continuously differentiable scalar functions on $\mathbb{R}^n$. If f has a local extremum at $\mathbf{c}$, subject to a constraint $g(\mathbf{x}) = k$, for $k \in \mathbb{R}$, and $\nabla g(\mathbf{c}) \neq 0$, then $\exists \lambda$ such that $\nabla f(\mathbf{c}) = \lambda \nabla g(\mathbf{c})$. This λ is known as the Lagrange multiplier.

A farmer has 200 metres of fencing with which to construct three sides of a rectangular pen, with an existing fence acting as the fourth side (fig. 3.3) What dimensions will maximise the area of the pen?

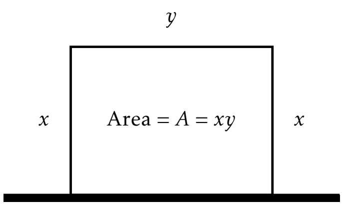

Figure 3.3: The farmer's pen

This obviously leads to the system of linear equations:

$$A(x,y) = xy$$
$$P(x,y) = 2x + y$$

However, $P(x,y)$ is subject to the constraint that $P(x,y) = 200$. So, we have $\nabla A(x,y) = (y,x)$ and $\nabla P(x,y) = (2,1)$. We arrive at a system of three equations in three unknowns

$$\begin{aligned} 2x + y &= 200 \\ y - 2\lambda &= 0 \\ x - \lambda &= 0 \end{aligned}$$

which corresponds to the augmented matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & 0 & 200 \\ 0 & 1 & -2 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right]$$

which reduces to[1]

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 50 \\ 0 & 1 & 0 & 100 \\ 0 & 0 & 1 & 50 \end{array}\right]$$

Thus, $x = 50$ and $y = 100$. This obviously was a trivial problem that could have been solved by a simple substitution, however the method of Lagrange multipliers proves useful in more complicated optimisation problems.

3.6 Taylor Series

It is often convenient to approximate a function by a polynomial; if it is too complicated to integrate, for example. Define

$$T_{n,a}(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

To be the *Taylor Polynomial* of $n'th$ degree of f around a. Then $f(x) = T_{n,a}(x) + R_{n,a}(x)$, where $R_{n,a}(x)$ is an *error term* or *remainder*. For example, the Taylor polynomial of $\sin x$ around 0:

$$T_{6,0}(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + R_{6,0}(x)$$

[1] refer: MATH1012 Mathematical Theory and Methods

The formular for $R_{n,a}(x)$ is given by

$$R_{n,a}(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-a)^{n+1}$$

for some $z \in [a, x]$. We do not know z, however it is possible to bound the error and thus know how accurate our polynomial expansions are[2]. For the expansion of $\sin x$ around 0,

$$R_{6,0} = \frac{f^{(7)}(z)}{7!}x^7 \quad z \in [0, x]$$

Lets say that we want to estimate $\sin(\pi/4)$:

$$|R_{6,0}(x)| = \frac{|\cos z|}{5040}|x|^7 \leq \frac{1}{5040}\left|\frac{\pi}{4}\right|^7 = 3.66 \times 10^{-5}$$

so we can be quite comfortable using our Taylor expansion up to 4 decimal places of $\sin(\pi/4)$. In fact

$$\sin(\pi/4) = \frac{\sqrt{2}}{2} \approx 0.707107 \quad \text{and} \quad R_{6,0}(\pi/4) = 0.707143$$

[2] refer: MATH1012 Mathematical Theory and Methods.